INTRODUCTION OF DATAMANAGEMENT

PROJECT REPORT

(Project Semester January-April 2025)

(COFFEE SHOP SALES)

Submitted by

Avnish Sharma

Registration No.-**12313239**

Programme and Section-**K23ER**

Course Code**: INT217-INTRODUCTION OF DATAMANAGEMENT**

Under the Guidance of

Vikas Mangotra

Discipline of CSE/IT

Lovely School of Computer Science and Engineering

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that **Avnish Sharma** bearing Registration no. **12313239** has completed **INT-217** project titled, **"Coffee Shop Sales"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Designation of the Supervisor

School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab.

Date: 11-04-2025

DECLARATION

I **Avnish Sharma**, student of Computer Science and Engineering under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 11-04-2025 Signature

Registration No. :- 12313239 Name of the student

Avnish Sharma

1. Introduction:-

In the dynamic and customer-driven landscape of the food and beverage industry, small businesses like coffee shops thrive not only on the quality of their products but also on how efficiently they analyze and respond to sales trends. In light of this, the development of a Coffee Shop Sales Dashboard serves as a critical tool for transforming raw transaction data into meaningful business insights.

This project centers around building a comprehensive dashboard to visualize and monitor the day-to-day sales performance of a coffee shop. The dataset used in this project includes detailed records of sales transactions such as Date of Sale, Product Name, Category (e.g., Coffee, Snacks, Beverages), Quantity Sold, Unit Price, Total Sale Value, Payment Method, and Customer Type (e.g., Walk-in or Online). By organizing and analyzing this information, the dashboard provides a clear snapshot of business operations.

The main objectives of the dashboard are:

- To track total sales and revenue generated over various periods (daily, weekly, monthly).
- To identify best-selling products and those that are underperforming.
- To analyze customer preferences based on product categories and purchase patterns.
- To monitor payment method trends, which can inform service improvements.
- To observe peak business hours or days, enabling better staffing and inventory decisions.

The dashboard employs interactive visuals such as bar graphs, pie charts, line graphs, and filterable tables, making it intuitive and accessible even for non-technical users. It empowers the coffee shop owner or manager to make data-driven decisions, such as adjusting inventory levels, offering discounts on slow-moving products, or developing targeted marketing campaigns.

By integrating this dashboard into daily operations, the coffee shop gains a valuable decision-support tool that enhances efficiency, boosts customer satisfaction, and ultimately improves profitability.

2. Datasets:-

The **Coffee Shop Sales** dataset from Maven Analytics provides a comprehensive view of transactions from a fictional coffee shop in New York City. Spanning from January to June, it encompasses 149,116 records across 11 fields, detailing aspects such as transaction dates, product categories, quantities sold, and payment methods. This dataset is instrumental for analyzing sales trends, customer behaviors, and operational efficiencies within the coffee shop environment.

LINK-:

https://mavenanalytics.io/data-playground?accessType=open&order=date_added%2Cdesc&search=Coffee%20Shop%20Sales

3. Datasets Preprocessing:-

To prepare the raw coffee shop sales data for dashboard visualization and in-depth analysis, several preprocessing steps were applied to enhance its structure, readability, and analytical potential. These transformations are reflected in the file and are summarized below:

1. Date and Time Feature Extraction

- **Month Name**: Extracted from transaction_date to allow month-wise sales comparisons.
- **Day Name**: Helps in analyzing sales by days of the week (e.g., Monday vs. Saturday).
- **Hour**: Extracted from transaction_time to identify peak sales hours.
- **Day of Week** and **Month**: Added as numerical values for filtering and time-series charts.

2. Sales Metrics Calculation

- **Total_bill**: A new column created by multiplying transaction_qty with unit_price to get the total amount for each transaction — crucial for revenue analysis.

3. Product Details Decomposition

- **Size** and **Trim**: Derived from the product_detail column, enabling granular breakdowns of product variants (e.g., "Large | Vanilla" split into size = "Large", trim = "Vanilla").

4. Column Reordering and Renaming (if applicable)

- Rearranged columns to follow a logical sequence from transaction details → product information → calculated features.
- Ensures better readability and user-friendliness in dashboards.

Final Outcome:-

The resulting **preprocessed dataset** is now optimized for dashboarding tools like **Excel,** with clean, enriched, and structured data that supports meaningful visual insights.

4. Analysis on dataset (for each objective)

I. General Description:-

The dataset provided from Maven Analytics – Coffee Shop Sales contains records of daily transactions in a coffee shop across three main locations:

- **Astoria**
- **Hell's Kitchen**
- **Lower Manhattan**

Each row in the dataset typically includes the following fields:

Field	Description
`transaction_id`	Unique identifier for each transaction
`date` & `time`	Timestamp of the transaction
`store_location`	Branch where transaction occurred
`product_category`	Category such as Coffee, Tea, Bakery, etc.
`product_type`	Specific product name like Espresso, Chai tea
`size`	Size of the item ordered: Small, Regular, Large
`unit_price`	Price per unit
`quantity`	Quantity sold
`total_price`	Total bill for the transaction
`customer_id`	Unique customer ID (used for footfall analysis)

ii. Specific Requirements:

The dashboard is designed to fulfill the following analysis objectives:

1. **Track overall performance** – Total sales, footfall, and average bill/order per person.
2. **Understand customer preferences** – Best-selling products, category distribution, and size preferences.
3. **Identify peak business hours and days**.
4. **Compare store performance** based on footfall and sales.
5. **Enable filtering by Month and Day** for dynamic reporting.

Objective	Insights
Overall Business Health	Total sales, average bill per customer, and average items per person
Peak Hour Analysis	Understand busiest hours to optimize staffing and inventory
Weekday Trends	Which days have the most orders or sales
Product Performance	Identify best-selling products
Category Contribution	What types of items (coffee, tea, etc.) drive most revenue
Size Preferences	Determine preferred sizes by customers
Store Comparison	Analyze footfall and revenue by location
Month/Day Filters	Allow deep-dive into specific time periods

iii. Analysis Results (with Examples)

Here's a more detailed breakdown of the key visuals and what they represent:

1. Quantity Based on Hours

- **Peak Hours:** 7 AM to 9 AM with over **900 units sold** during this timeframe.
- Indicates **high morning demand**, aligns with typical coffee consumption behavior.

2. Category % Distribution (by Sales)

- **Coffee**: 38% – clearly the top revenue driver.
- **Tea**: 27% – strong secondary category.
- Other notable contributors:
  - Drinking Chocolate: 12%
  - Coffee Beans: 10%
  - Bakery: 6%

3. % Size Distribution (by Orders)

- **Large**: 38%
- **Regular**: 30%
- **Small**: 9%
- **Not Defined**: 23% (likely missing/uncategorized data to be cleaned).

4. Footfall Over Store Locations

- **Hell's Kitchen**: Highest footfall & revenue.
- Lower Manhattan and Astoria follow.
- Used: COUNT of customer_id for footfall & SUM of total_price for sales.

5. Top 5 Products (Based on Sales)

Product	**Sales**
Barista Espresso	$2,499.15
Brewed Chai Tea	$2,005.85
Gourmet Brewed Coffee	$1,871.00
Hot Chocolate	$1,856.75
Brewed Herbal Tea	$1,287.00

6. Orders on Weekdays

- **Tuesday & Wednesday** have highest **sales and customer visits**.
- Sundays have lowest footfall and sales.

iv. Visualization Techniques

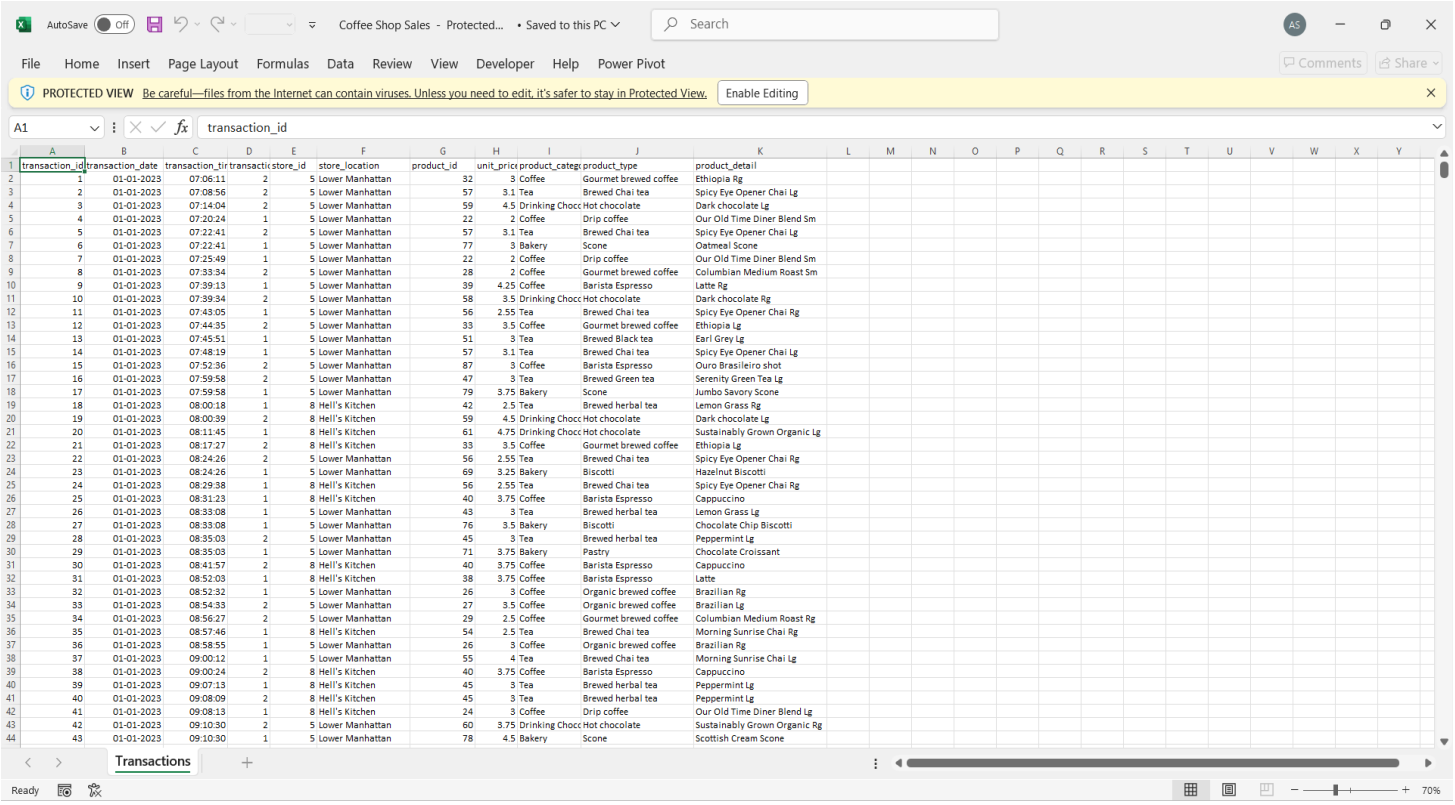
All visuals were created using **Excel Pivot Tables** and **Pivot Charts**. Key formatting & preprocessing steps include:

Excel File Differences & Preprocessing (Your File vs Original)

Step	Changes Made in Format.xlsx
Cleaning	Removed unnecessary columns, corrected NULL values, especially in size and product_type
DateTime Split	Split combined date_time field into Date and Hour
Text Formatting	Standardized capitalization for Store Name, Product Category, and Size
Pivot Tables	Created separate pivot tables for each visual (total sales, category-wise sales, footfall, etc.)
Slicers/Filters	Added Month and Day Name slicers for interactive filtering
Formatting	Consistent color theme (light brown), icon usage, shadow effects for modern dashboard look
Custom Calculations	Calculated fields like:

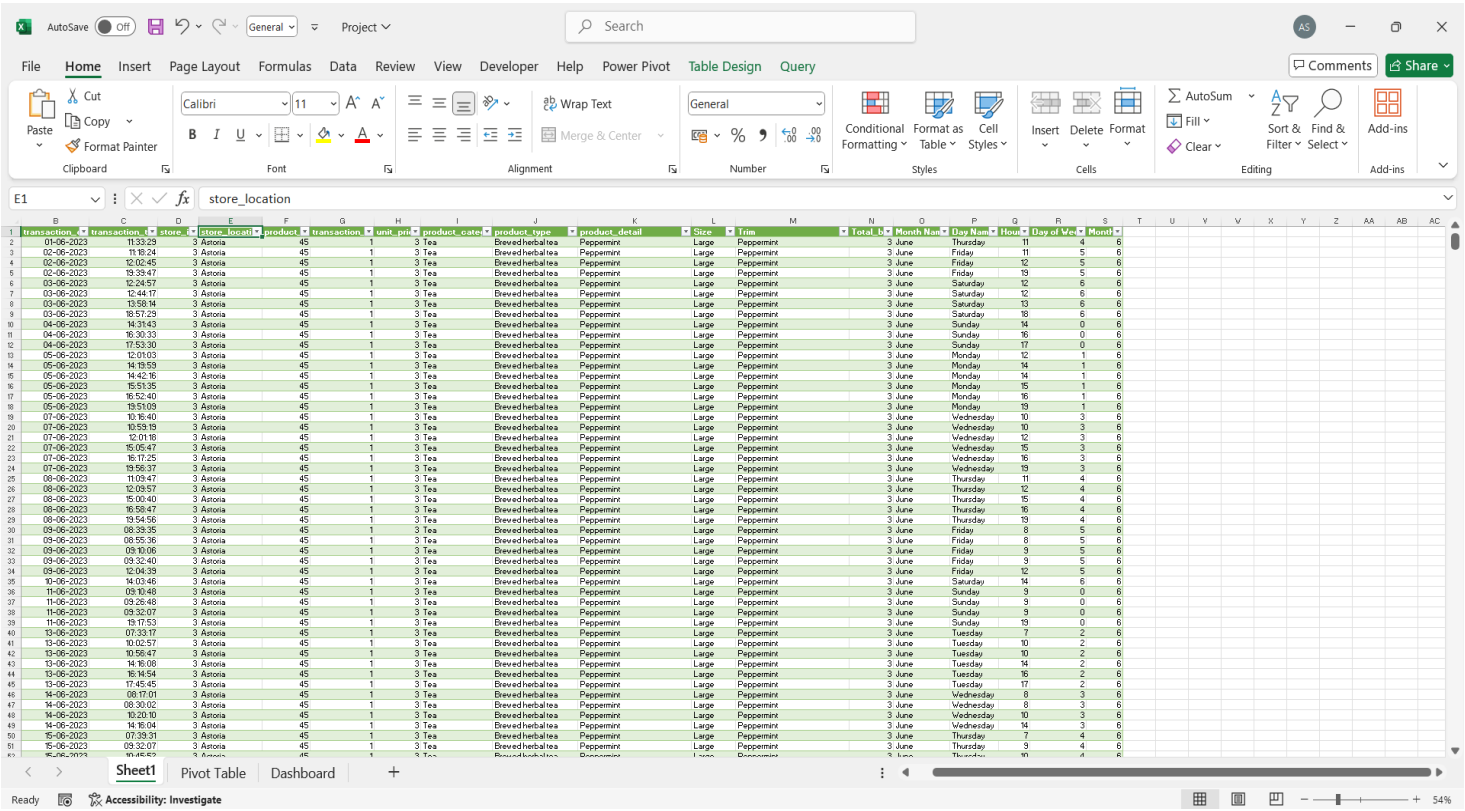
- Avg Bill/Person = Total Sales / Footfall
- Avg Order/Person = Total Quantity / Footfall

Raw Data Of Excel:-



transaction_id	transaction_date	transaction_tir	transactic	store_id	store_location	product_id	unit_price	product_categ	product_type	product_detail
1	01-01-2023	07:06:11	2	5	Lower Manhattan	32	3	Coffee	Gourmet brewed coffee	Ethiopia Rg
2	01-01-2023	07:08:56	2	5	Lower Manhattan	57	3.1	Tea	Brewed Chai tea	Spicy Eye Opener Chai Lg
3	01-01-2023	07:14:04	2	5	Lower Manhattan	59	4.5	Drinking Chocc	Hot chocolate	Dark chocolate Lg
4	01-01-2023	07:20:24	1	5	Lower Manhattan	22	2	Coffee	Drip coffee	Our Old Time Diner Blend Sm
5	01-01-2023	07:22:41	2	5	Lower Manhattan	57	3.1	Tea	Brewed Chai tea	Spicy Eye Opener Chai Lg
6	01-01-2023	07:22:41	1	5	Lower Manhattan	77	3	Bakery	Scone	Oatmeal Scone
7	01-01-2023	07:25:49	1	5	Lower Manhattan	22	2	Coffee	Drip coffee	Our Old Time Diner Blend Sm
8	01-01-2023	07:33:34	2	5	Lower Manhattan	28	2	Coffee	Gourmet brewed coffee	Columbian Medium Roast Sm
9	01-01-2023	07:39:13	1	5	Lower Manhattan	39	4.25	Coffee	Barista Espresso	Latte Rg
10	01-01-2023	07:39:34	2	5	Lower Manhattan	58	3.5	Drinking Chocc	Hot chocolate	Dark chocolate Rg
11	01-01-2023	07:43:05	1	5	Lower Manhattan	56	2.55	Tea	Brewed Chai tea	Spicy Eye Opener Chai Rg
12	01-01-2023	07:44:35	2	5	Lower Manhattan	33	3.5	Coffee	Gourmet brewed coffee	Ethiopia Lg
13	01-01-2023	07:45:51	1	5	Lower Manhattan	51	3	Tea	Brewed Black tea	Earl Grey Lg
14	01-01-2023	07:48:19	1	5	Lower Manhattan	57	3.1	Tea	Brewed Chai tea	Spicy Eye Opener Chai Lg
15	01-01-2023	07:52:36	2	5	Lower Manhattan	87	3	Coffee	Barista Espresso	Ouro Brasileiro shot
16	01-01-2023	07:59:58	2	5	Lower Manhattan	47	3	Tea	Brewed Green tea	Serenity Green Tea Lg
17	01-01-2023	07:59:58	1	5	Lower Manhattan	79	3.75	Bakery	Scone	Jumbo Savory Scone
18	01-01-2023	08:00:18	1	8	Hell's Kitchen	42	2.5	Tea	Brewed herbal tea	Lemon Grass Rg
19	01-01-2023	08:00:39	2	8	Hell's Kitchen	59	4.5	Drinking Chocc	Hot chocolate	Dark chocolate Lg
20	01-01-2023	08:11:45	1	8	Hell's Kitchen	61	4.75	Drinking Chocc	Hot chocolate	Sustainably Grown Organic Lg
21	01-01-2023	08:17:27	2	8	Hell's Kitchen	33	3.5	Coffee	Gourmet brewed coffee	Ethiopia Lg
22	01-01-2023	08:24:26	2	5	Lower Manhattan	56	2.55	Tea	Brewed Chai tea	Spicy Eye Opener Chai Rg
23	01-01-2023	08:24:26	1	5	Lower Manhattan	69	3.25	Bakery	Biscotti	Hazelnut Biscotti
24	01-01-2023	08:29:38	1	8	Hell's Kitchen	56	2.55	Tea	Brewed Chai tea	Spicy Eye Opener Chai Rg
25	01-01-2023	08:31:23	1	8	Hell's Kitchen	40	3.75	Coffee	Barista Espresso	Cappuccino
26	01-01-2023	08:33:08	1	5	Lower Manhattan	43	3	Tea	Brewed herbal tea	Lemon Grass Lg
27	01-01-2023	08:33:08	1	5	Lower Manhattan	76	3.5	Bakery	Biscotti	Chocolate Chip Biscotti
28	01-01-2023	08:35:03	2	5	Lower Manhattan	45	3	Tea	Brewed herbal tea	Peppermint Lg
29	01-01-2023	08:35:03	1	5	Lower Manhattan	71	3.75	Bakery	Pastry	Chocolate Croissant
30	01-01-2023	08:41:57	2	8	Hell's Kitchen	40	3.75	Coffee	Barista Espresso	Cappuccino
31	01-01-2023	08:52:03	1	8	Hell's Kitchen	38	3.75	Coffee	Barista Espresso	Latte
32	01-01-2023	08:52:32	1	5	Lower Manhattan	26	3	Coffee	Organic brewed coffee	Brazilian Rg
33	01-01-2023	08:54:33	2	5	Lower Manhattan	27	3.5	Coffee	Organic brewed coffee	Brazilian Lg
34	01-01-2023	08:56:27	2	5	Lower Manhattan	29	2.5	Coffee	Gourmet brewed coffee	Columbian Medium Roast Rg
35	01-01-2023	08:57:46	1	8	Hell's Kitchen	54	2.5	Tea	Brewed Chai tea	Morning Sunrise Chai Rg
36	01-01-2023	08:58:55	1	5	Lower Manhattan	26	3	Coffee	Organic brewed coffee	Brazilian Rg
37	01-01-2023	09:00:12	1	5	Lower Manhattan	55	4	Tea	Brewed Chai tea	Morning Sunrise Chai Lg
38	01-01-2023	09:00:24	2	8	Hell's Kitchen	40	3.75	Coffee	Barista Espresso	Cappuccino
39	01-01-2023	09:07:13	1	8	Hell's Kitchen	45	3	Tea	Brewed herbal tea	Peppermint Lg
40	01-01-2023	09:08:09	2	8	Hell's Kitchen	45	3	Tea	Brewed herbal tea	Peppermint Lg
41	01-01-2023	09:08:13	1	8	Hell's Kitchen	24	3	Coffee	Drip coffee	Our Old Time Diner Blend Lg
42	01-01-2023	09:10:30	2	5	Lower Manhattan	60	3.75	Drinking Chocc	Hot chocolate	Sustainably Grown Organic Rg
43	01-01-2023	09:10:30	1	5	Lower Manhattan	78	4.5	Bakery	Scone	Scottish Cream Scone

After Data Preprocessing:-



transaction_	transaction_d	store_	store_locati	product_	transaction_	unit_pri	product_cate	product_type	product_detail	Size	Tim	Total_b	Month Nar	Day Nam	Hour	Day of We	Month
01-06-2023	11:33:29	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Thursday	11	4	6
02-06-2023	11:18:24	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Friday	11	5	6
02-06-2023	12:02:45	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Friday	12	5	6
02-06-2023	19:39:47	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Friday	19	5	6
03-06-2023	12:24:57	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Saturday	12	6	6
03-06-2023	12:44:17	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Saturday	12	6	6
03-06-2023	13:58:14	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Saturday	13	6	6
03-06-2023	18:57:28	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Saturday	18	6	6
04-06-2023	14:31:43	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Sunday	14	0	6
04-06-2023	16:30:33	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Sunday	16	0	6
04-06-2023	17:53:30	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Sunday	17	0	6
05-06-2023	12:01:03	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Monday	12	1	6
05-06-2023	14:19:59	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Monday	14	1	6
05-06-2023	14:42:16	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Monday	14	1	6
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05-06-2023	16:52:40	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Monday	16	1	6
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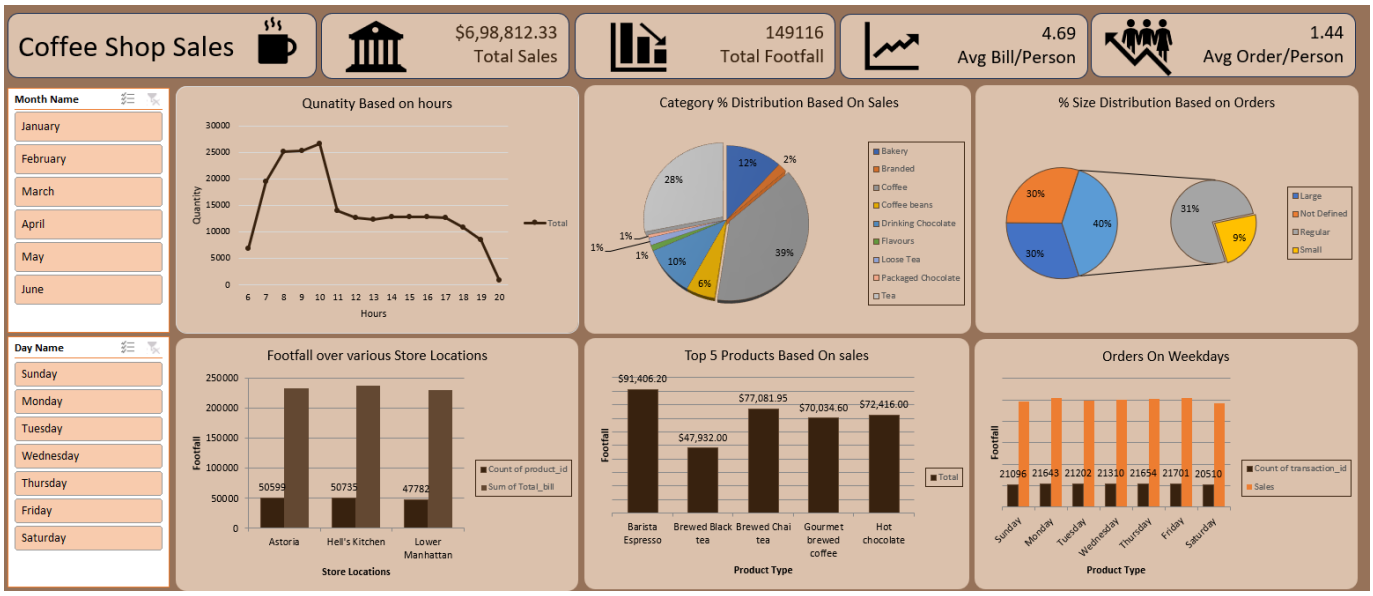
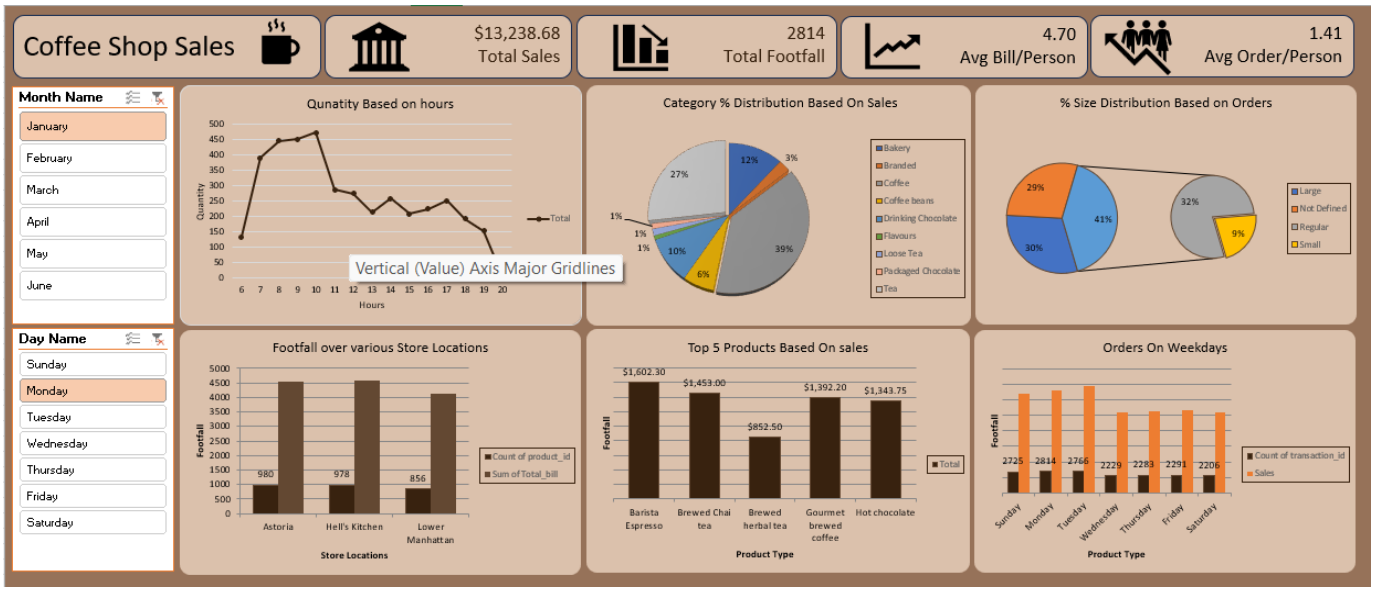
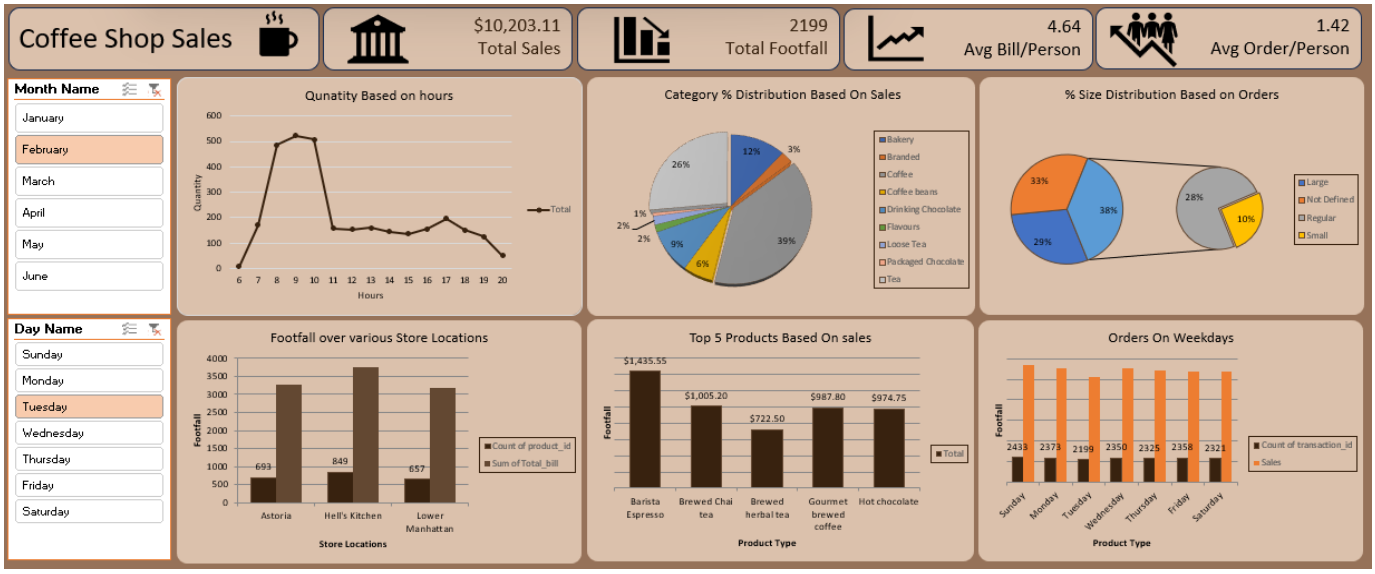
07-06-2023	10:16:40	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Wednesday	10	3	6
07-06-2023	10:59:19	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Wednesday	10	3	6
07-06-2023	12:01:18	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Wednesday	12	3	6
07-06-2023	15:05:47	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Wednesday	15	3	6
07-06-2023	16:17:25	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Wednesday	16	3	6
07-06-2023	19:56:37	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Wednesday	19	3	6
08-06-2023	11:09:47	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Thursday	11	4	6
08-06-2023	12:09:57	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Thursday	12	4	6
08-06-2023	15:00:40	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Thursday	15	4	6
08-06-2023	16:58:47	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Thursday	16	4	6
08-06-2023	19:54:56	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Thursday	19	4	6
09-06-2023	08:39:35	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Friday	8	5	6
09-06-2023	08:55:36	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Friday	8	5	6
09-06-2023	09:10:06	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Friday	9	5	6
09-06-2023	09:32:40	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Friday	9	5	6
09-06-2023	12:04:39	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Friday	12	5	6
10-06-2023	14:03:46	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Saturday	14	6	6
11-06-2023	09:10:48	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Sunday	9	0	6
11-06-2023	09:26:48	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Sunday	9	0	6
11-06-2023	09:32:07	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Sunday	9	0	6
11-06-2023	19:17:53	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Sunday	19	0	6
13-06-2023	07:33:17	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Tuesday	7	2	6
13-06-2023	10:02:57	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Tuesday	10	2	6
13-06-2023	10:56:47	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Tuesday	10	2	6
13-06-2023	14:16:08	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Tuesday	14	2	6
13-06-2023	16:14:54	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Tuesday	16	2	6
13-06-2023	17:45:45	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Tuesday	17	2	6
14-06-2023	08:17:01	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Wednesday	8	3	6
14-06-2023	08:30:02	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Wednesday	8	3	6
14-06-2023	10:20:10	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Wednesday	10	3	6
14-06-2023	14:16:04	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Wednesday	14	3	6
15-06-2023	07:39:31	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Thursday	7	4	6
15-06-2023	09:32:07	3	Astoria	45	1	3	Tea	Brewed herbal tea	Peppermint	Large	Peppermint	3	June	Thursday	9	4	6

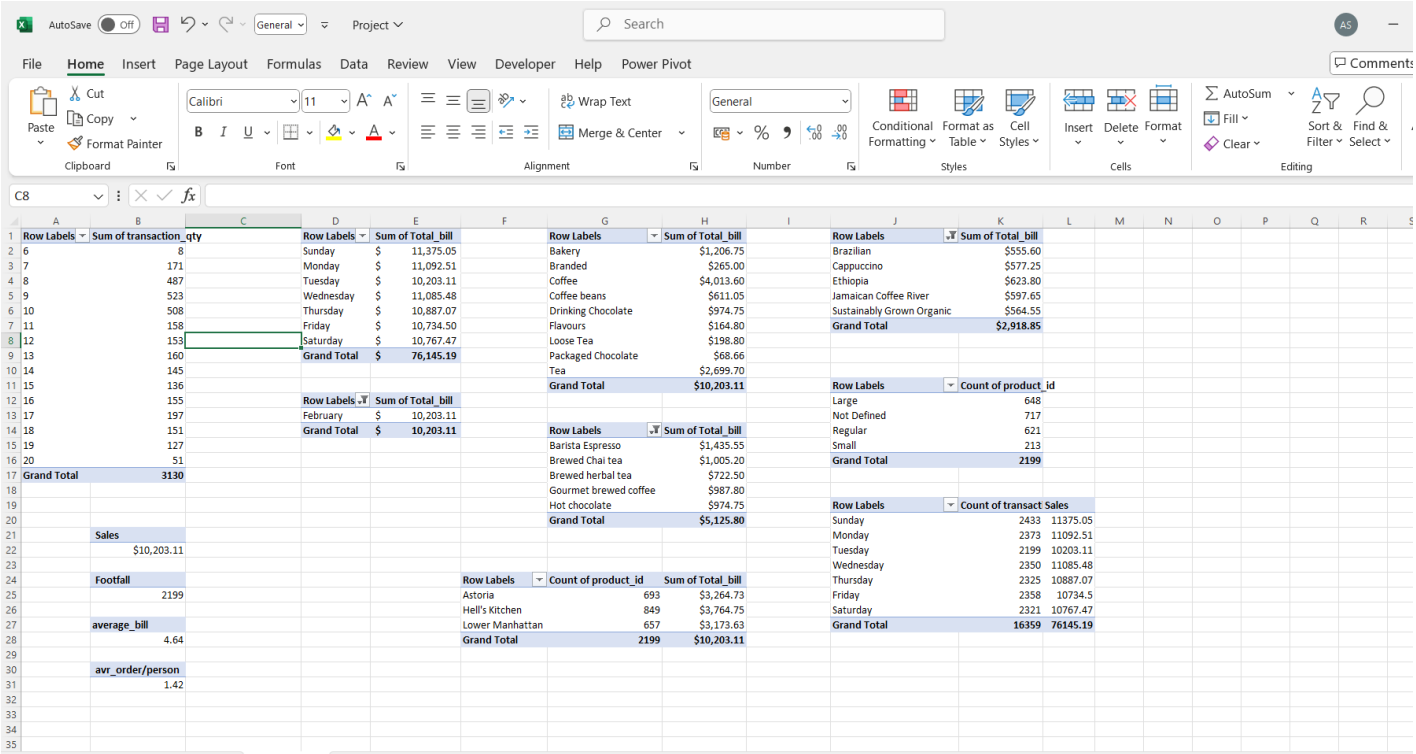
Dashboard:-


Coffee Shop Sales
$6,98,812.33
Total Sales
149116
Total Footfall
4.69
Avg Bill/Person
1.44
Avg Order/Person
Month Name
January
February
March
April
May
June
Day Name
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Qunatity Based on hours
Quantity
Hours
Total
Category % Distribution Based On Sales
Bakery
Branded
Coffee
Coffee beans
Drinking Chocolate
Flavours
Loose Tea
Packaged Chocolate
Tea
% Size Distribution Based on Orders
Large
Not Defined
Regular
Small
Footfall over various Store Locations
Footfall
Astoria
Hell's Kitchen
Lower Manhattan
Store Locations
Count of product_id
Sum of Total_bill
Top 5 Products Based On sales
$91,406.20
$47,932.00
$77,081.95
$70,034.60
$72,416.00
Barista Espresso
Brewed Black tea
Brewed Chai tea
Gourmet brewed coffee
Hot chocolate
Product Type
Orders On Weekdays
21096
21643
21202
21310
21654
21701
20510
Count of transaction_id
Sales


Coffee Shop Sales
$13,238.68
Total Sales
2814
Total Footfall
4.70
Avg Bill/Person
1.41
Avg Order/Person
Month Name
January
February
March
April
May
June
Day Name
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Qunatity Based on hours
Vertical (Value) Axis Major Gridlines
Category % Distribution Based On Sales
% Size Distribution Based on Orders
Footfall over various Store Locations
980
978
856
Top 5 Products Based On sales
$1,602.30
$1,453.00
$852.50
$1,392.20
$1,343.75
Barista Espresso
Brewed Chai tea
Brewed herbal tea
Gourmet brewed coffee
Hot chocolate
Orders On Weekdays
2725
2814
2766
2229
2283
2291
2206


Coffee Shop Sales
$10,203.11
Total Sales
2199
Total Footfall
4.64
Avg Bill/Person
1.42
Avg Order/Person
Month Name
January
February
March
April
May
June
Day Name
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Qunatity Based on hours
Category % Distribution Based On Sales
% Size Distribution Based on Orders
Footfall over various Store Locations
693
849
657
Top 5 Products Based On sales
$1,435.55
$1,005.20
$722.50
$987.80
$974.75
Barista Espresso
Brewed Chai tea
Brewed herbal tea
Gourmet brewed coffee
Hot chocolate
Orders On Weekdays
2433
2373
2199
2350
2325
2358
2321

Pivot Table:-



Row Labels	Sum of transaction_qty
6	8
7	171
8	487
9	523
10	508
11	158
12	153
13	160
14	145
15	136
16	155
17	197
18	151
19	127
20	51
Grand Total	3130

Sales
$10,203.11

Footfall
2199

average_bill
4.64

avr_order/person
1.42

Row Labels	Sum of Total_bill
Sunday	$ 11,375.05
Monday	$ 11,092.51
Tuesday	$ 10,203.11
Wednesday	$ 11,085.48
Thursday	$ 10,887.07
Friday	$ 10,734.50
Saturday	$ 10,767.47
Grand Total	$ 76,145.19

Row Labels	Sum of Total_bill
February	$ 10,203.11
Grand Total	$ 10,203.11

Row Labels	Sum of Total_bill
Bakery	$1,206.75
Branded	$265.00
Coffee	$4,013.60
Coffee beans	$611.05
Drinking Chocolate	$974.75
Flavours	$164.80
Loose Tea	$198.80
Packaged Chocolate	$68.66
Tea	$2,699.70
Grand Total	$10,203.11

Row Labels	Sum of Total_bill
Barista Espresso	$1,435.55
Brewed Chai tea	$1,005.20
Brewed herbal tea	$722.50
Gourmet brewed coffee	$987.80
Hot chocolate	$974.75
Grand Total	$5,125.80

Row Labels	Count of product_id	Sum of Total_bill
Astoria	693	$3,264.73
Hell's Kitchen	849	$3,764.75
Lower Manhattan	657	$3,173.63
Grand Total	2199	$10,203.11

Row Labels	Sum of Total_bill
Brazilian	$555.60
Cappuccino	$577.25
Ethiopia	$623.80
Jamaican Coffee River	$597.65
Sustainably Grown Organic	$564.55
Grand Total	$2,918.85

Row Labels	Count of product_id
Large	648
Not Defined	717
Regular	621
Small	213
Grand Total	2199

Row Labels	Count of transact	Sales
Sunday	2433	11375.05
Monday	2373	11092.51
Tuesday	2199	10203.11
Wednesday	2350	11085.48
Thursday	2325	10887.07
Friday	2358	10734.5
Saturday	2321	10767.47
Grand Total	16359	76145.19

Linkdin link:-

https://www.linkedin.com/in/geekyavnishsharma/recent-activity/all/

5. Conclusion

The Coffee Shop Sales Dashboard provides a comprehensive overview of business performance, customer behavior, and product trends. With total sales amounting to $698,812.33 and a footfall of 149,116 customers, the average bill per person stands at $4.69 with approximately 1.44 orders per customer. Coffee and Drinking Chocolate emerged as the most popular categories, contributing 39% and 28% to overall sales respectively, followed by Tea at 12%. Barista Espresso is the top-selling product, generating $91,406.20 in revenue, followed closely by Brewed Chai Tea and Hot Chocolate.

Sales data indicates that the busiest hours are between 7 AM and 9 AM, underscoring strong commuter-driven demand. In contrast, sales see a significant drop during the late afternoon hours from 5 PM to 8 PM, suggesting an opportunity to implement time-based promotions like afternoon discounts or loyalty rewards. A major observation from the order data is that 31% of the orders are marked as “Not Defined” in terms of size, highlighting the need for better data entry practices and potential refinement in the point-of-sale (POS) system. Among all store locations, Hell’s Kitchen leads in both footfall and total sales, indicating its strategic advantage and possibly higher visibility or accessibility. Lower Manhattan also performs strongly, while

Astoria reports the lowest performance, hinting at the need for targeted community outreach or promotional strategies in underperforming areas. Day-wise analysis reveals Tuesday and Wednesday as the most active days for transactions, while Sunday records the least, indicating an opportunity to introduce weekend specials, family combos, or localized events to boost footfall.

Overall, the dashboard proves to be a valuable tool for identifying operational strengths and areas of improvement. It enables informed decision-making through clear, data-backed insights. Implementing the recommended strategies, such as enhancing data quality, launching weekday and morning-specific campaigns, and re-engaging customers during off-peak hours and low-traffic days, can lead to increased efficiency, customer satisfaction, and overall revenue growth.

6. Future scope

- The Coffee Shop Sales Dashboard sets a strong foundation for data-driven decision-making, and its scope can be further expanded to unlock deeper insights and greater business value. One key area for future development is the integration of real-time data from POS systems, which would allow store managers and corporate teams to monitor sales trends live and respond proactively to changes in customer behavior.

- Predictive analytics could also be introduced to forecast demand based on historical data, seasonal patterns, and external factors such as weather or local events. This would help optimize inventory management and staff scheduling, reducing waste and improving customer service.

- Another potential enhancement is implementing customer segmentation and loyalty tracking. By analyzing individual purchase histories, preferences, and visit frequencies, personalized marketing campaigns and reward programs can be designed to increase customer retention and satisfaction.

- The dashboard could be extended to include cost and profit analysis, offering insights into product margins and operational expenses. This would support better pricing strategies and resource allocation.

- In addition, expanding the dashboard to include mobile and cloud accessibility would ensure managers and executives can make informed decisions from anywhere, enhancing agility and responsiveness.

- Finally, improving data entry practices and automating data collection can ensure higher accuracy, which is critical for effective reporting and forecasting. By embracing these innovations, the coffee shop chain can stay competitive, scale efficiently, and deliver an enhanced customer experience.

7. References

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